

# CALC BOY

## User Manual



Version 3.1.0 - Generated from the current repository state.

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# Introduction

CALC BOY is a handheld-console-style calculator. This manual teaches the everyday workflows, explains when to use each page, and shows the results you should expect.

Tip: The display is interactive. Tap it for history; long-press it to copy the current result.

## Installation

- Open the live link in Safari or Chrome.
- Use Share or the browser menu and choose Add to Home Screen.
- Launch CALC BOY from the home screen. Installed mode opens the app fullscreen.

### Offline Mode

After the first online launch, the browser keeps the app available for later. CALC BOY can then open without a connection as long as the site's stored data remains on the device.

Tip: After the first load, CALC BOY can open even without an internet connection.

## First calculation



Example:  $12 + 30 =$  shows 42 and adds the expression to history.

# History, copy and share



The last ten successful calculations are saved. The history view also shows sum and average. Text and PNG export are available when the browser supports sharing or clipboard APIs.

# BASIC



Standard calculator for everyday arithmetic, signs, square roots and smart percent calculations.

## Workflow

- Step 1** Enter the first number, choose an operator, enter the second number, then press =.
- Step 2** Use AC to clear the current calculation. After ERROR, the next digit starts a new entry.
- Step 3** Tap the LCD to open history; long-press the LCD to copy the displayed result.

Good to know: The percent key is context-aware: with + or - it calculates a percentage of the first operand.

## Examples

### Smart percent

**Input / action**  
100 + 10 % = 110; 100 - 10 %  
**Result**  
90

### Square root

**Input / action**  
144 sqrt  
**Result**  
12

## Button reference

| Key  | Function                                                             | Key | Function                                                                    |
|------|----------------------------------------------------------------------|-----|-----------------------------------------------------------------------------|
| AC   | Clears entry and active operation; in RPN it also clears the stack.  | *   | Multiplies two values.                                                      |
| +/-  | Changes the sign of the displayed number.                            | 1   | Digit input.                                                                |
| %    | Smart percent. Example: 100 + 10 % becomes 110.                      | 2   | Digit input.                                                                |
| sqrt | Calculates the square root; negative values show ERROR.              | 3   | Digit input.                                                                |
| 7    | Digit input.                                                         | -   | Subtracts the second value from the first.                                  |
| 8    | Digit input.                                                         | 0   | Digit input.                                                                |
| 9    | Digit input.                                                         | ,   | Adds the decimal separator.                                                 |
| /    | Divides the first value by the second; division by zero shows ERROR. | +   | Adds two values.                                                            |
| 4    | Digit input.                                                         | EXT | Opens the extended scientific page.                                         |
| 5    | Digit input.                                                         | =   | Evaluates the current calculation; in RPN it pushes the value to the stack. |
| 6    | Digit input.                                                         |     |                                                                             |

# EXT



Scientific and memory page with trigonometry, logarithms, powers, roots, factorial, pi and e.

## Workflow

- Step 1** For direct functions such as sin, log,  $x^2$  or  $1/x$ , enter a value and press the function key.
- Step 2** For  $x^y$ , enter the base, press  $x^y$ , enter the exponent, then press =.
- Step 3** Use DEG/RAD before trigonometry. DEG is saved and is the default unless changed.

Good to know: Factorial works only for whole numbers from 0 to 170.

## Examples

### DEG/RAD

#### Input / action

DEG: 30 sin -> 0.5; RAD: pi sin

#### Result

approximately 0

### Memory

#### Input / action

42 M+, AC, MR

#### Result

42; 10 M- leaves 32 in memory

### Powers

#### Input / action

$2 \times^y 8 =$

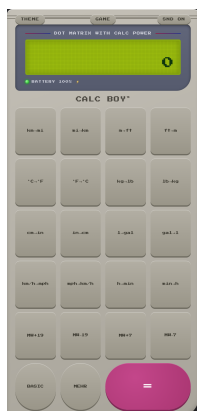
#### Result

256

## Button reference

| Key     | Function                                               | Key   | Function                                                                    |
|---------|--------------------------------------------------------|-------|-----------------------------------------------------------------------------|
| MC      | Clears calculator memory.                              | $x^2$ | Squares the displayed value.                                                |
| MR      | Recalls the saved memory value to the display.         | $x^3$ | Cubes the displayed value.                                                  |
| M+      | Adds the displayed value to memory.                    | $x^y$ | Binary power: enter base, press $x^y$ , enter exponent, press =.            |
| M-      | Subtracts the displayed value from memory.             | cbt   | Calculates the cube root.                                                   |
| sin     | Sine of the displayed value, using DEG or RAD mode.    | $1/x$ | Calculates the reciprocal; 0 shows ERROR.                                   |
| cos     | Cosine of the displayed value, using DEG or RAD mode.  | n!    | Calculates factorial for whole numbers from 0 to 170.                       |
| tan     | Tangent of the displayed value, using DEG or RAD mode. | pi    | Inserts pi.                                                                 |
| DEG/RAD | Toggles saved angle mode for trigonometry.             | e     | Inserts Euler's number e.                                                   |
| log     | Base-10 logarithm of the displayed value.              | BASIC | Returns to the standard calculator page.                                    |
| In      | Natural logarithm of the displayed value.              | MEHR  | Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.                           |
| $10^x$  | Calculates ten to the power of the displayed value.    | =     | Evaluates the current calculation; in RPN it pushes the value to the stack. |
| $e^x$   | Calculates e to the power of the displayed value.      |       |                                                                             |

# CONV



One-tap converters for distance, temperature, mass, volume, speed, time and German VAT.

## Workflow

- Step 1 Enter the source value and press the desired conversion key.
- Step 2 Use the matching reverse key if you need to convert back.
- Step 3 VAT keys treat the entered value as net or gross according to the key label.

Good to know: US gallons are used for litre/gallon conversion.

## Examples

### VAT

**Input / action**  
100 MW+19 -> 119; 119 MW-19  
**Result**  
100

### Temperature

**Input / action**  
20 C->F -> 68; 68 F->C  
**Result**  
20

### Speed

**Input / action**  
100 km/h->mph  
**Result**  
62.137119224

## Button reference

| Key    | Function               | Key       | Function                                                                    |
|--------|------------------------|-----------|-----------------------------------------------------------------------------|
| km->mi | Kilometres to miles.   | km/h->mph | Kilometres per hour to miles per hour.                                      |
| mi->km | Miles to kilometres.   | mph->km/h | Miles per hour to kilometres per hour.                                      |
| m->ft  | Metres to feet.        | h->min    | Hours to minutes.                                                           |
| ft->m  | Feet to metres.        | min->h    | Minutes to hours.                                                           |
| C->F   | Celsius to Fahrenheit. | MW+19     | Adds 19 percent German VAT to a net amount.                                 |
| F->C   | Fahrenheit to Celsius. | MW-19     | Removes 19 percent German VAT from a gross amount.                          |
| kg->lb | Kilograms to pounds.   | MW+7      | Adds 7 percent German VAT to a net amount.                                  |
| lb->kg | Pounds to kilograms.   | MW-7      | Removes 7 percent German VAT from a gross amount.                           |
| cm->in | Centimetres to inches. | BASIC     | Returns to the standard calculator page.                                    |
| in->cm | Inches to centimetres. | MEHR      | Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.                           |
| l->gal | Litres to US gallons.  | =         | Evaluates the current calculation; in RPN it pushes the value to the stack. |
| gal->l | US gallons to litres.  |           |                                                                             |



Finance helpers for compound interest with monthly savings, interest amount, tips, bill splitting and manual currency conversion.

## Workflow

- Step 1 Set K0, P%, years and monthly rate with SET keys. ENDWERT calculates the future value.
- Step 2 ZINSEN shows only the earned interest: future value minus start capital and deposits.
- Step 3 Set persons before / PERS. Set exchange rate before EUR->\$ or \$->EUR.

Good to know: Default values are K0 1000, P 3, years 10, monthly rate 50, persons 2 and exchange rate 1.08.

## Examples

### Compound interest

**Input / action**  
SET K0=1000, SET P%=3, SET JAHRE=10, SET RATE/M=50,  
ENDWERT  
**Result**  
8336.42449101

### Interest only

**Input / action**  
With the same values, ZINSEN  
**Result**  
1336.42449101

### Bill split

**Input / action**  
SET PERS=4, enter 80, / PERS  
**Result**  
20

### Currency

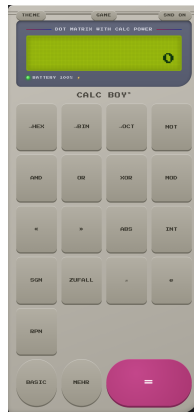
**Input / action**  
SET KURS=1.08, 100 EUR->\$ -> 108; 108 \$->EUR  
**Result**  
100

Important: Finance values remain stored until RESET or site data is cleared.

## Button reference

| Key        | Function                                                              | Key      | Function                                                                    |
|------------|-----------------------------------------------------------------------|----------|-----------------------------------------------------------------------------|
| SET K0     | Stores start capital for compound interest.                           | TIP+20%  | Adds 20 percent tip to the displayed amount.                                |
| SET P%     | Stores yearly interest rate in percent.                               | SET PERS | Stores number of people for bill splitting.                                 |
| SET JAHRE  | Stores duration in years.                                             | / PERS   | Divides the displayed amount by the stored number of people.                |
| SET RATE/M | Stores the monthly savings amount.                                    | SET KURS | Stores the manual exchange rate.                                            |
| INFO       | Shows stored parameters or variables.                                 | EUR->\$  | Converts euros to dollars with the stored rate.                             |
| ENDWERT    | Calculates future value including start capital and monthly payments. | \$->EUR  | Converts dollars to euros with the stored rate.                             |
| ZINSEN     | Shows only earned interest.                                           | BASIC    | Returns to the standard calculator page.                                    |
| RESET      | Restores finance defaults.                                            | MEHR     | Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.                           |
| TIP+10%    | Adds 10 percent tip to the displayed amount.                          | =        | Evaluates the current calculation; in RPN it pushes the value to the stack. |
| TIP+15%    | Adds 15 percent tip to the displayed amount.                          |          |                                                                             |

# PRG



Programmer functions, integer base display, bitwise operators, random number and RPN mode.

## Workflow

- Step 1      HEX, BIN and OCT show the integer part of the display in the status line.
- Step 2      AND, OR, XOR, MOD, << and >> are binary operators: enter first value, operator, second value, =.
- Step 3      RPN changes = into push: enter a value, press =, enter the next value, then press an operator.

Good to know: Only the integer part of the number is used for bit operations and base display.

## Examples

### Base display

**Input / action**  
255 ->HEX shows FF  
**Result**  
BIN shows 11111111

### Binary operations

**Input / action**  
5 AND 3 = 1; 5 OR 2 = 7; 9 MOD 4  
**Result**  
1

### RPN

**Input / action**  
RPN on: 12 =, 3 +  
**Result**  
15; AC clears the stack

Important: RPN changes how = behaves: it pushes a value onto the stack instead of finishing a normal calculation.

## Button reference

| Key   | Function                                                  | Key    | Function                                                                    |
|-------|-----------------------------------------------------------|--------|-----------------------------------------------------------------------------|
| ->HEX | Shows the integer part in hexadecimal in the status line. | ABS    | Absolute value.                                                             |
| ->BIN | Shows the integer part in binary in the status line.      | INT    | Integer part without decimals.                                              |
| ->OCT | Shows the integer part in octal in the status line.       | SGN    | Sign of the number: -1, 0 or 1.                                             |
| NOT   | Bitwise NOT of the integer part.                          | ZUFALL | Random integer from 1 to 100.                                               |
| AND   | Bitwise AND for two integer operands.                     | pi     | Inserts pi.                                                                 |
| OR    | Bitwise OR for two integer operands.                      | e      | Inserts Euler's number e.                                                   |
| XOR   | Bitwise XOR for two integer operands.                     | RPN    | Toggles Reverse Polish Notation mode.                                       |
| MOD   | Integer remainder; modulo by 0 shows ERROR.               | BASIC  | Returns to the standard calculator page.                                    |
| <<    | Shifts the integer part left.                             | MEHR   | Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.                           |
| >>    | Shifts the integer part right.                            | =      | Evaluates the current calculation; in RPN it pushes the value to the stack. |



# PLOT



Draws 20 built-in function graphs as pixel plots on the LCD.

## Workflow

- Step 1 Open PLOT and press a function key such as sin x,  $x^2$  or GAUSS.
- Step 2 The graph replaces the LCD temporarily and scales itself automatically.
- Step 3 Press any other key or tap the display to close the graph.

Good to know: Asymptotes such as tan x or  $1/x$  are clipped so the plot stays readable.

## Examples

sin x

**Input / action**  
PLOT -> sin x  
**Result**  
 $y = \sin x$

Compare

**Input / action**  
PLOT ->  $x^2$  or  $x^3$   
**Result**  
automatic scaling

## Button reference

| Key    | Function                                          | Key              | Function                                                                    |
|--------|---------------------------------------------------|------------------|-----------------------------------------------------------------------------|
| sin x  | Draws $y = \sin x$ .                              | sinh             | Draws hyperbolic sine.                                                      |
| cos x  | Draws $y = \cos x$ .                              | cosh             | Draws hyperbolic cosine.                                                    |
| tan x  | Draws $y = \tan x$ with clipping near asymptotes. | tanh             | Draws hyperbolic tangent.                                                   |
| $x^2$  | Draws $y = x^2$ .                                 | $x^4$            | Draws $y = x^4$ .                                                           |
| $x^3$  | Draws $y = x^3$ .                                 | GAUSS            | Draws the Gaussian bell curve $e^{(-x^2)}$ .                                |
| sqrt x | Draws $y = \text{square root of } x$ .            | FLOOR            | Draws the floor step function.                                              |
| log x  | Draws $y = \log_{10} x$ .                         | sinc x           | Draws $\sin x / x$ , with value 1 at $x = 0$ .                              |
| $1/x$  | Calculates the reciprocal; 0 shows ERROR.         | $x \cdot \sin x$ | Draws $y = x \text{ times } \sin x$ .                                       |
| $e^x$  | Draws $y = e^x$ .                                 | BASIC            | Returns to the standard calculator page.                                    |
| ln x   | Draws $y = \ln x$ .                               | MEHR             | Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.                           |
| x      | Draws the absolute-value graph.                   | =                | Evaluates the current calculation; in RPN it pushes the value to the stack. |
| $2^x$  | Draws $y = 2^x$ .                                 |                  |                                                                             |

# FORM



Formula assistant based on three stored variables A, B and C.

## Workflow

- Step 1** Enter a number and press SET A, SET B or SET C to store it.
- Step 2** Press INFO to check stored values. CLR ABC resets all variables to 0.
- Step 3** Press a formula key; the result appears on the display and is added to history.

Good to know: Formula keys use fixed roles: for example BMI uses A as kg and B as metres; speed uses A as kilometres and B as hours.

## Examples

### Percent

#### Input / action

SET A=20, SET B=150, % VON

#### Result

30

### Pythagoras

#### Input / action

SET A=3, SET B=4, PYTH

#### Result

5

### BMI

#### Input / action

SET A=80, SET B=1.8, BMI

#### Result

24.6913580247

### Net/gross

#### Input / action

SET A=119, NET19 -> 100; SET A=100, BRU19

#### Result

119

## Button reference

| Key      | Function                                  | Key     | Function                                                                    |
|----------|-------------------------------------------|---------|-----------------------------------------------------------------------------|
| SET A    | Stores the displayed value in variable A. | BMI     | BMI with A as kg and B as metres.                                           |
| SET B    | Stores the displayed value in variable B. | AVG ABC | Average of A, B and C.                                                      |
| SET C    | Stores the displayed value in variable C. | CLR ABC | Resets A, B and C to 0.                                                     |
| INFO     | Shows stored parameters or variables.     | KM/H    | Speed: A kilometres divided by B hours.                                     |
| % VON    | Calculates A percent of B.                | NET19   | Net amount from a gross amount with 19 percent VAT.                         |
| DREISATZ | Calculates B times C divided by A.        | BRU19   | Gross amount from a net amount with 19 percent VAT.                         |
| KREIS A  | Circle area with radius A.                | BASIC   | Returns to the standard calculator page.                                    |
| KREIS U  | Circle circumference with radius A.       | MEHR    | Cycles EXT -> CONV -> FIN -> PRG -> PLOT -> FORM.                           |
| PYTH     | Pythagoras: square root of $A^2 + B^2$ .  | =       | Evaluates the current calculation; in RPN it pushes the value to the stack. |
| OHM      | Ohm helper: A times B.                    |         |                                                                             |

# Plotting



Draws 20 built-in function graphs as pixel plots on the LCD.

## Workflow

- Step 1 | Open PLOT and press a function key such as  $\sin x$ ,  $x^2$  or GAUSS.
- Step 2 | The graph replaces the LCD temporarily and scales itself automatically.
- Step 3 | Press any other key or tap the display to close the graph.

## Examples

$\sin x$

### Input / action

PLOT ->  $\sin x$

### Result

$y = \sin x$

Compare

### Input / action

PLOT ->  $x^2$  or  $x^3$

### Result

automatic scaling

Tip: Press any non-plot key or tap the display to close the graph.

# Games



Math Attack asks arithmetic questions for 30 seconds. Snake uses the number pad 2/4/6/8 or arrow keys and stores its own high score.

## Landscape mode



In landscape, the scientific shortcut column is visible beside BASIC, giving quick access to sin, cos, tan, log, ln, powers, pi and e.

## Keyboard

| Key        | Action                                |
|------------|---------------------------------------|
| 0-9        | digits                                |
| , or .     | decimal separator                     |
| + - * /    | basic operators                       |
| Enter or = | equals                                |
| Escape     | AC                                    |
| %          | percent                               |
| r or w     | square root                           |
| arrow keys | Snake direction and secret-code input |

## Storage and Privacy

Privacy: No usage data is sent to external analytics or font services.

All settings stay on your device. CALC BOY does not use analytics, does not load external fonts, and does not fetch live exchange rates.

theme, sound, angle mode, history, memory, finance parameters, exchange rate, persons, RPN mode, RPN stack, formula variables, game high scores, Virtual Boy unlock.

## Limitations

Offline use requires a first launch from the website. Currency conversion uses the manually stored rate and does not fetch live rates. Programmer base conversion appears in the status line and does not replace the main display. Sharing, clipboard, vibration and battery display depend on browser support. The in-app interface is German.

## Troubleshooting

If an old version appears after update, close and reopen the app twice. If sound does not play, tap once inside the app and check SND ON. If sharing is unavailable, use clipboard copy or browser download fallback. To reset all data, delete website data for this domain.

## Version Information

CALC BOY 3.1.0 - calcboy-v3.1.0 - GPL-3.0-or-later

# Project information

Links, license and version at a glance.

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Project website                | <a href="https://schrotty74.github.io/CalcBoy/">https://schrotty74.github.io/CalcBoy/</a>               |
| GitHub repository              | <a href="https://github.com/Schrotty74/CalcBoy">https://github.com/Schrotty74/CalcBoy</a>               |
| License                        | GPL-3.0-or-later                                                                                        |
| Report bugs / request features | <a href="https://github.com/Schrotty74/CalcBoy/issues">https://github.com/Schrotty74/CalcBoy/issues</a> |
| Current version                | 3.1.0                                                                                                   |

Good to know: Use the project website for the live app. The GitHub repository contains the source code and GPL-3.0-or-later license; use Issues for bug reports and feature requests.